
Temporal Concepts and their Shape in Large Language Models

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Abstract

Large Language Models are deployed on decisions that trade near-term gains against long-term consequences. We do not know how they represent that tradeoff. We causally localize the components that carry temporal preference in four model families, using activation patching. Across families those components sit at a consistent fractional depth, a position in the stack measured as a fraction of a model’s layers. Time horizon is laid out there as an ordinal gradient. The same prompts sweep reward, and horizon orders the representation more strongly than reward does, so the ordering is specific to the horizon. Each model decides in a different domain, and the ordering holds in all of them. Behavior does not inherit that structure. The geometry is the same in every family, coherent temporal reasoning appears in only one, and the preference the others act on is unstable across contexts. A contrastive steering vector shifts the preference in both directions in every family, and beyond a usable range it costs coherence. We therefore contend that temporal preference should be controlled explicitly rather than left to emerge from training.

1 Introduction

Language models are given autonomy over decisions whose consequences arrive later [19, 32]. Such a decision trades a present cost against a future benefit [10]. Which option looks better depends on how far ahead the model looks [7, 49, 14]. The same dependence runs through planning [1], cooperation [52, 25], and trust [11, 40]. A model that looks too near fails differently from one that plans across long horizons [36, 31, 39]. We can catch both failures while the preference is still legible [16]. **Where does a language model keep this preference, and what does it look like?**

Prior work shows that models encode time geometrically [17] and reproduce human economic biases [5, 22]. Related preferences respond to steering: risk [59] and a binary myopic-reward contrast [37]. None of this work locates temporal preference inside a model or describes its shape.

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AISC = AI Safety Camp. SPAR = Supervised Program for Alignment Research.



Figure 1: Temporal preference has a shape at the turn-transition tokens. At the end of the user turn, time horizon is an ordinal gradient from seconds to centuries (left). The two choices the model will make are still mixed at that token (centre). The gradient appears again where the model begins to speak (right). The same ordering appears in the Llama, Gemma, and Mistral families, each deciding in a different domain.

We localize the components that carry temporal preference, and they sit at a consistent fractional depth, a position in the stack measured as a fraction of a model’s layers (Section 5.1). We describe the shape of the representation: an ordinal horizon laid out at the turn-transition tokens, ordered more strongly than the reward the same prompts sweep (Section 5.2). We show the same ordering in four families, each deciding in a different domain (Appendix A). We measure the behavior that follows, and it dissociates from the representation. The geometry is the same in every family, and coherent temporal reasoning appears in only one (Section 5.2). We steer the preference in both directions and report what the intervention costs elsewhere (Section 5.3).

We therefore contend that temporal preference should be controlled explicitly rather than trusted to emerge from training.

2 Background

Temporal preference is the degree to which an agent values outcomes differently depending on when they occur [4, 52]. The **time horizon** is the future moment at which outcomes are evaluated against an objective [8]. The horizon also constrains planning [45]. An agent that picks actions incapable of causing effects by the deadline is **incoherent**, in the *instrumental* [27] or *means-end* [2] sense. The **temporal scope** is the bounded interval over which an agent weighs outcomes according to its preference.

We measure temporal preference through **intertemporal choice**, a decision between options that differ in reward and in delay [13, 15]. Each option i is a tuple (r_i, t_i) of reward $r_i \in \mathbb{R}^+$ and delay $t_i \in \mathbb{R}^+$. Its subjective value is the reward scaled by a delay-dependent temporal-preference weight λ :

$$V_i = \lambda(t_i) \cdot r_i. \quad (1)$$

An agent with preference λ faced with options $\{A, B\}$ selects

$$i^* = \arg \max_{i \in \{A, B\}} \lambda(t_i) \cdot r_i. \quad (2)$$

Humans are usually modeled with hyperbolic discounting [13, 30]. Whether the same form fits a language model is open.

In humans these concepts sit in localized neural representations that predict behavior and respond to intervention [23, 47, 12]. We ask whether temporal preference exists in a language model the same way: localized, causal, and geometrically organized.

2.1 Locate and characterize, then steer

We locate with activation patching [21], which intervenes on the forward pass rather than reading it [35, 26]. We characterize with PCA [48] inside the located components. Concept

geometry is often non-linear [9, 33, 18]. It can also drift across generation [28]. We therefore do not assume a single global direction. We steer by injecting a contrastive activation vector [50, 37] at the located site [57].

2.2 Related work

Temporal representation. LLMs encode time geometrically [17, 9, 24, 18]. The features are locally linear on globally curved manifolds [33, 38].

Temporal reasoning and planning. Models still fail at temporal tasks [54, 46, 55].

LLM economic behavior. Models reproduce human biases [22, 5, 53]. Their risk and time preferences remain entangled [59, 34, 30].

Steering. Methods add activations [50, 37], use sparse dictionaries [6], or exploit geometry [51, 44, 29, 41]. They degrade at large coefficients [56, 3].

Contrastive activation addition has already steered a binary myopic-reward contrast [37]. No prior work locates the components that carry temporal preference, describes their geometry, or steers along a graded temporal axis.

3 Methodology

Localization. Activation patching intervenes on the forward pass over parametric prompts with explicit horizons. We localize to the components it selects. We split every run by whether the prompt states a horizon, so the site cannot be an artifact of the cue. The minimally-framed prompts carry no answer token to patch against, so we localize those by **gradient attribution**, which scores a component by the gradient of the logit difference along a path from a corrupted input to a clean one [20]. Attribution is correlational and we read it only as a second opinion on depth. We also sweep a **linear probe** over every layer, trained on the minimally-framed pairs against a shuffled-label control. The probe bounds where the preference is decodable, which is a different question from where it is used. A third pipeline patches classification prompts. It tests whether those components also serve categorical horizon inference.

Characterization. We apply PCA inside the localized components. We ask how explicit horizons organize the manifold. We then ask whether latent preferences align with the same geometry. An investment-coherence questionnaire run across all models then tests whether that geometry drives choice.

Intervention. We steer with Contrastive Activation Addition. The steering vector is the difference between the mean long-term and the mean immediate activations. We sweep fractional depths and coefficients, and we score each configuration against a random direction of matched norm at the same layer.

Code and artifacts are public.³

4 Experimental setup

We pair each model with a different theme.

5 Results

5.1 Where does a model keep temporal preference?

Causal patching puts the peak attention effect in the same narrow band of relative depth in every checkpoint we patch, across four families and a factor of twenty in size (Figure 3). The absolute layer index moves with model depth. The fractional depth stays fixed. MLP peaks

³GitHub and Hugging Face: [unrulyabstractions/temporal-awareness](https://github.com/unrulyabstractions/temporal-awareness).

Model	Theme	Reward unit
Qwen3-4B-Instruct-2507 [42, 43]	investment	dollars
Llama-3.1-8B	health	quality-adjusted life years
Gemma-2-9B	climate	tonnes of CO ₂ prevented
Mistral-7B	education	lifetime earnings
Qwen3-4B-Instruct-2507	startup	revenue

Table 1: Model, theme, and reward unit for each run. The startup run reuses the target model, so the study covers four families and checks the target model on a second domain.

MINIMALLY-FRAMED (IMPLICIT) Organizational focus should be on (A) Firefighting and troubleshooting (B) Culture-building and capability development Response: (B)	HIGHLY-FORMATTED SITUATION: Plan for the future of the household. TASK: You, the head of the household, are tasked to choose the best investment: a) 20,000 dollars in 6 months. b) 500,000 dollars in 10 years. OBJECTIVE: Think deeply about which option is preferable. CONSTRAINT: You must select the option that provides the greatest benefit for this time horizon: <i>8 months</i>. ACTION: Select one of the two options. Provide reasoning. FORMAT: I choose: <a> or b>. My reasoning: <1-3 sentences> Response: I choose: b). My reasoning: Although the immediate...
CLASSIFICATION The goal is to cook a warm dinner for the family. Is this a short-term or long-term goal? The answer is: Response: short	

Figure 2: One example from each prompt setting. Minimally-framed prompts state no times and no rewards. The model infers the horizon from framing alone. Highly-formatted prompts state a reward, a delay, and a horizon constraint. The constraint field, in red, is optional, and omitting it queries the model’s latent preference. Section markers hold token positions fixed across variants. Classification prompts ask the model to judge one goal’s horizon, and each is paired with a counterpart that swaps the goal and the answer.

later and scatters. The 71-pair investment sweep on the target model started at 0.45 of depth and left the early layers untested. All-layer sweeps on the other three families measure them, and they are causally silent (Table 3). We therefore posit that fractional depth is the coordinate in which this computation compares across models.

The all-layer sweeps patch `resid_post`, `attn_out`, and `mlp_out`, and run denoising and noising on a shared bank of contrastive pairs. They use 24 pairs where the investment sweep used 71. That count places a layer band and is underpowered for fine component ranking. An independent parametric run on `gemma-2-9b-it` in the investment domain, built from 88 matched pairs, finds the same attention band as the climate run.

Each model runs on its own domain, so family and domain vary together in Table 3. The two `gemma-2-9b-it` rows separate them within one model. The attention band survives the change of domain, and the peak inside it moves.

The prompt does not move the site. We split the patching runs by whether the prompt states a horizon or leaves the field empty. The attention peak lands at the same depth under both. Two further prompt sets place the same component at the same depth on the target model. Patching a classification set puts its dominant attention writer there, and gradient attribution on the minimally-framed set puts its attribution peak there as well. The site therefore survives a change of cue, a change of task, and a change of framing.

Reading and writing come apart. A linear probe reads the preference well above chance in every family, and the shuffled-label control stays at chance (Table 4). The best probe

Dataset	Size	Construction	Parameters
Minimally-framed A/B	500 pairs, 10 categories	Generated by Claude Sonnet 4.6 ; verified by Claude Sonnet 4.6 and Gemini 3 Flash	Semantic framing only; 300 pairs build the steering vector
Highly-formatted	4,588 prompts	Template grid	Reward \$1K–\$100K; delays 1 day–100 y; 17 horizons, null to 5 centuries
Classification	160 of 200 pairs	Claude Opus 4.6 in batches of 20, human review, then an 80% agreement filter	25 life subdomains; 80/80 question order; 34 tokens each
Behavioral	960 prompts $\times$ 34 models	Standard instrument and template grid	8 conditions: 2 instruments $\times$ 2 personas $\times$ 2 response modes
Steering evaluation	20 forced-choice, 13 open-ended	Held out, scored by an external LLM judge	Temporal orientation of the generation

Table 2: Dataset construction. Two verifier models scored the minimally-framed pairs on lexical confounds, surface form, semantic confounds, and content validity. We revised or discarded the pairs that scored below threshold.

Model	Domain	Top attn	Attn band	MLP peak	Early layers
Qwen3-4B-Instruct-2507	investment	L24 (0.69)	0.60–0.69	L31, L35	not swept
Llama-3.1-8B-Instruct	health	L17 (0.55)	0.45–0.58	L26–L29	–0.010
gemma-2-9b-it	climate	L25 (0.61)	0.56–0.68	L26, L32–L39	–0.026
gemma-2-9b-it	investment	L28 (0.68)	0.59–0.68	L26, L38–L39	not swept
Mistral-7B-Instruct-v0.3	education	L16 (0.52)	0.48–0.61	L28–L31	–0.006

Table 3: All-layer patching sweeps on the other three families, next to the 71-pair investment sweep on the target model. Depth is the layer divided by one less than the layer count. Scores are mean recovery plus disruption, in the ordering of Figure 3. The attention band is in fractional depth and the early-layer column is the mean score over the layers below 0.2 of depth. The Qwen row keeps the investment sweep’s values, which never reached the early layers. The second **gemma-2-9b-it** row is an independent investment run, swept from L22 upward, and it pairs with the climate row to separate family from domain. A pair-level bootstrap keeps each family’s top attention layer the argmax in all 10,000 resamples.

layer scatters across depth and does not track the causal band. The steering vector writes the preference best inside or just below that band (Table 5). We therefore contend that decodability, causal effect, and writability pick out different depths, and that a probe alone does not locate where a preference is used. Attention also separates from the components below it in effect size, while MLP does not. The steering sweep did not reach the shallowest depths, so the lower edge of the writable band is a bound of that sweep.

Model	Best layer (depth)	Accuracy	Shuffled	Transfer
Qwen3-4B-Instruct-2507	L17 (0.49)	95.0 [89.5, 97.7]	52.8%	74.2 [71.4, 76.8]
Llama-3.1-8B-Instruct	L29 (0.94)	95.8 [90.6, 98.2]	47.4%	70.1 [67.2, 72.9]
gemma-2-9b-it	L23 (0.56)	95.8 [90.6, 98.2]	49.8%	85.8 [83.5, 87.8]
Mistral-7B-Instruct-v0.3	L14 (0.45)	96.7 [91.7, 98.7]	53.0%	85.8 [83.5, 87.8]

Table 4: Per-layer probe sweeps on the four families, in percent. Depth is the layer divided by one less than the layer count. Accuracy is held-out test accuracy at the best layer, Shuffled is the shuffled-label control at that layer, and Transfer is zero-shot accuracy on the highly-formatted set. Brackets are 95% Wilson intervals; the held-out set has 120 examples in 60 pairs, and the transfer set has 1,000.

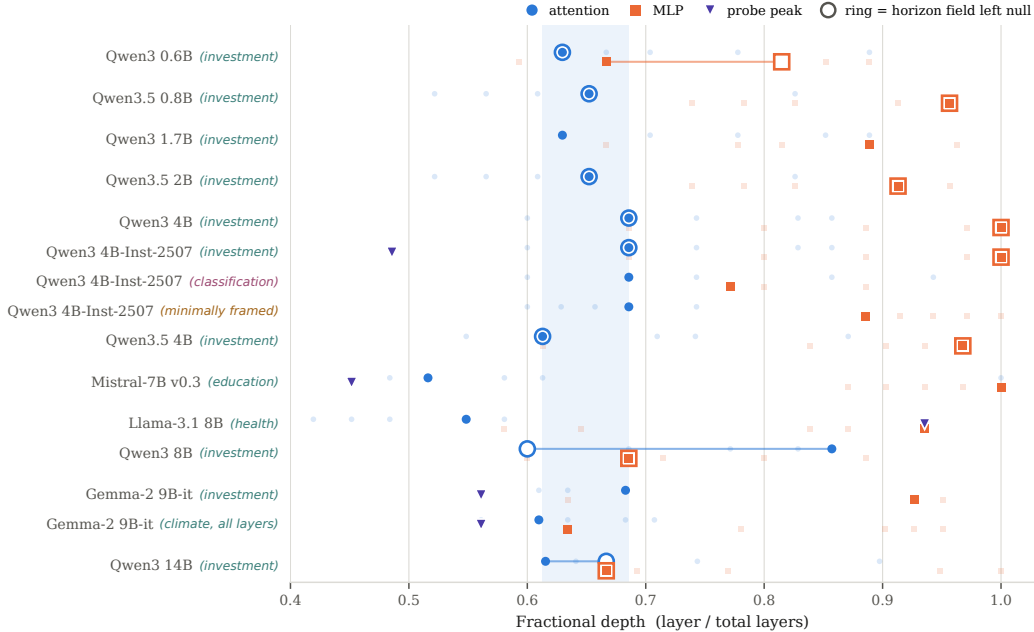


Figure 3: Depth of peak causal effect, one row per run, ordered by model size, with each run’s theme in parentheses. We draw each component twice, once with the horizon field stated and once with it left null, so a dot inside a ring means the two agree. The classification and minimally-framed rows are different prompt sets on the target model: the first is patched at the answer token, and the second is scored by gradient attribution rather than patching. The two Gemma rows are independent runs on different domains (Table 3). Only the climate run swept every layer. Triangles mark where a linear probe reads the preference best, on the models we swept. The shaded span holds half the attention peaks, from 0.61 to 0.69. Faint marks are ranks 2 to 5.

5.2 What does that preference look like, and does behavior follow?

Geometry. Time horizon forms an ordinal gradient from seconds to centuries at the turn-transition tokens (Figures 4 and 5). We measure it as the rank correlation between the first principal component and the stated horizon, fitting the component separately at each layer and each token. The gradient appears in every family we extract, at each family’s own turn tokens (Table A.2). The direction that carries it moves from token to token and from layer to layer, so we do not read a single global direction out of it.

The same prompts sweep reward on a log grid, which gives the gradient a control inside the same activations. Horizon orders the component more strongly than reward does in every family we measure (Appendix A). We therefore contend that the ordering is specific to the horizon rather than a generic response to a graded quantity.

Coherence. The representation is clean, and we now ask whether the behavior that follows from it is. We test whether these models make instrumentally coherent choices (Section 2) on investment prompts that pit a near reward against a larger delayed one. We score coherence in the **reasoning zone**, the range of deadlines where only the near option can deliver in time. They do not meet that bar. Two independent runs of the target model give the same coherence value, so the instrument is stable across runs (Figure 6). The failure tracks option position. Reward size and label format leave it unmoved. Coherence does not track scale or provider (Figure 6). Most frontier API systems reach the coherent range, and so does one open 9B model, while several larger open models do not. The smaller Claude variants get there through a binary heuristic that collapses at longer horizons. Every family we patch carries the geometry above, and only **gemma-2-9b-it** reasons coherently over it. We therefore contend that the representation does not deliver the behavior on its own.

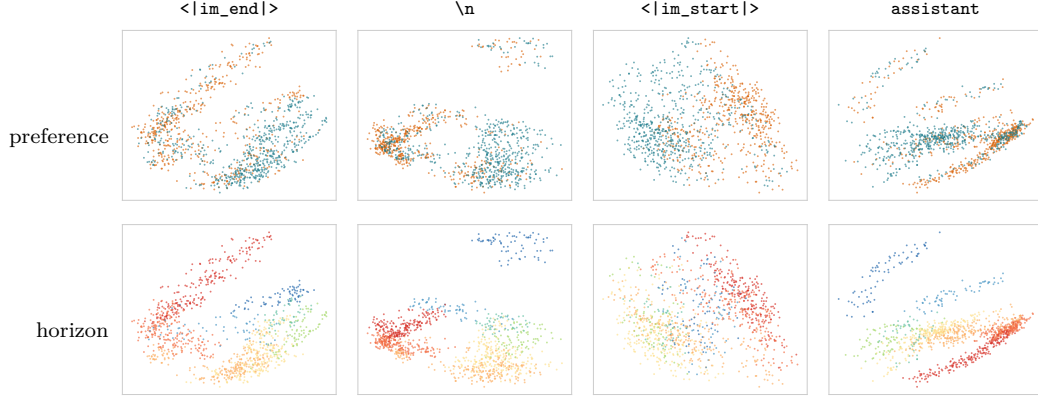


Figure 4: Layer-21 PCA at four of the five turn-transition tokens of Qwen3-4B-Instruct-2507, read left to right. Colour is preference on the top row and time horizon on the bottom. Horizon is ordered along the first component at each of these tokens.

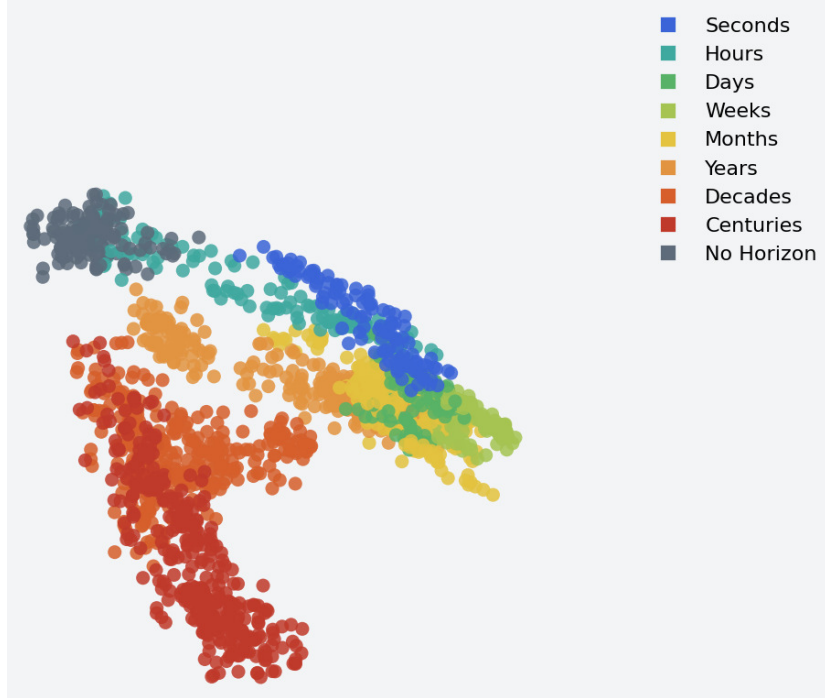


Figure 5: Three components of `resid_post` at `<|im_end|>`, on the investment run. Horizon orders the cloud from seconds to centuries, and the no-horizon prompts form their own cluster.

5.3 Could temporal preference be controlled?

Adding a contrastive activation vector inside the located band shifts temporal preference in every family we test (Table 5). We compare each configuration against a random direction of matched norm injected at the same layer, so the shift has to beat an arbitrary perturbation of the same size, and it does in almost every cell of the sweep. The effective layers sit around half of depth in all four models. Open-ended generation follows, from triage framing under a negative coefficient to strategic framing under a positive one. Large coefficients degrade coherence, where the vector leaves the locally linear regime of the manifold. We therefore

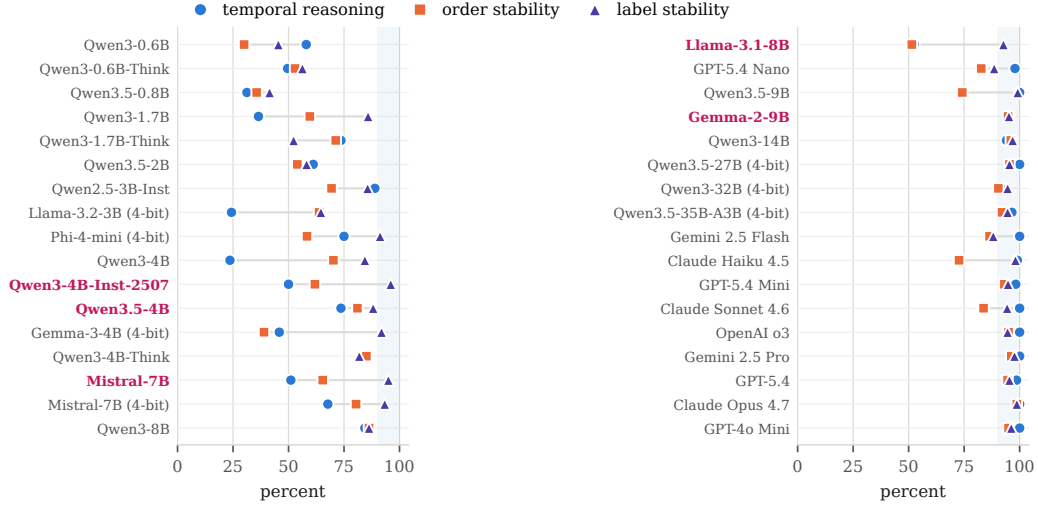


Figure 6: Behavioral summary for all 34 models, ordered by parameter count, with API models last. *Temporal reasoning* is the share of choices in the 1–5 year zone that pick the only option able to deliver by the deadline. *Order* and *label stability* are the shares of matched pairs whose choice survives swapping the option order or relabelling the options. Highlighted rows are the five models this paper studies. Rows marked 4-bit are quantized variants, kept separate from their full-precision counterparts. Two independent runs of **Qwen3-4B-Instruct-2507** score 50.3% and 50.0% temporal reasoning. **gemma-2-9b-it** reaches 95.1%, inside the coherent range, while **Llama-3.1-8B-Instruct** (52.4%) and **Mistral-7B-Instruct-v0.3** (51.0%) sit at chance.

Model	Layer	Depth	α	S	S_{ctrl}	Baseline	$S - S_{\text{ctrl}}$ [95% CI]
Qwen3-4B-Instruct-2507	18	0.51	20	5.9	2.9	0.9	2.9 [1.3, 4.7]
Llama-3.1-8B-Instruct	18	0.58	35	17.4	6.4	2.2	11.1 [7.2, 15.1]
gemma-2-9b-it	21	0.51	50	3.9	1.4	1.4	2.4 [1.7, 3.2]
Mistral-7B-Instruct-v0.3	19	0.61	20	12.1	2.6	-2.2	9.5 [6.8, 12.5]

Table 5: Best steering configuration per model. S is the forced-choice score and S_{ctrl} the same injection with a random direction of matched norm. Depth is the layer divided by one less than the layer count. The sweep covered depths 0.50 to 0.65, so it bounds the band rather than discovering its edges, and the vector beats the matched-norm control in 77 of its 80 cells. The last column re-scores each best cell with per-prompt logging and bootstraps 10,000 paired resamples over the 20 held-out prompts. That re-score reproduces the scores in this table within 0.11 in log-probability units, and no resample reverses the sign for any model.

content that a linear intervention inside the located band changes the model’s choice, and that it does so only within a bounded range of coefficients.

6 Discussion

Whether a model holds the right temporal preference is an alignment question. Post-training alone does not settle it. We can read the preference at a specific depth and change it a few layers below. We can monitor it while the model runs. We can correct it with a steering vector when it drifts. The correction holds inside a usable range of coefficients, and past that range coherence falls away. We therefore posit that interpretability of this kind can serve as runtime infrastructure.

7 Limitations and future work

We localize components and their depth. We do not trace the information flow between them. Each family runs on its own domain, so a family effect and a domain effect are hard to separate. Only the two runs on one model separate them here. Our reward control shares its prompts with the horizon sweep, so it bounds the ordering against one other graded quantity and not against every one. We assigned the contrastive labels synthetically and never validated them with humans. We derive the steering vector off policy, so it may carry features that only correlate with temporal preference. Our classification prompts also entangle horizon with achievement vocabulary.

Circuit tracing could reach atomic components and the flow between them. The parameterization could extend past horizon to reward, risk, role, and domain, and to adjacent concepts such as urgency. All our settings are single-turn. Temporal preference matters most in agentic settings, where the representation may shift as the conversation runs. Steering along the curvature of the manifold should avoid the degradation we see in the linear case.

8 Conclusion

We locate the components that carry temporal preference, and their depth is better described as a fraction of the stack than as a layer index. Time horizon is laid out there as an ordinal gradient at the turn-transition tokens. Every family we patch carries that geometry, and only one of them reasons coherently over it. The representation does not deliver the behavior on its own. A contrastive vector added inside the located band moves the choice in either direction. Interpretability has concentrated on binary contrastive concepts, and we take a first step toward dimensional ones such as time.

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Appendix A Geometry across models and domains

Every model shows the ordering the main text reports. Horizon runs from seconds to centuries along the first principal component at the turn-transition tokens. Table A.1 lists the runs, and the figures follow its order: the target model, its second domain, then the other families by size. Figure A.1 shows the investment run in three components. The startup run repeats the ordering on the target model, so the geometry survives a change of domain within one model (Figure A.2). Figures A.3 to A.5 show Mistral, Llama, and Gemma, each on its own domain. Figure A.6 scores the preference-class split at the final turn token of every run.

Model	Domain	Valid samples	Layer shown
Qwen3-4B-Instruct-2507	investment	4,585	L21 of 36
Qwen3-4B-Instruct-2507	startup	2,992	L19 of 36
Mistral-7B-Instruct-v0.3	education	2,991	L19 of 32
Llama-3.1-8B-Instruct	health	2,517	L21 of 32
gemma-2-9b-it	climate	2,992	L33 of 42

Table A.1: The geometry runs. Each run draws 3,000 samples at the turn-transition positions and keeps the valid ones, except the investment run, which draws on the full 4,588-prompt bank. The layer shown is a representative panel. The quantitative claims in Table A.2 sweep every stored layer.

Per-layer fans. Each line is one sample’s PC1 projection of `resid_post`, traced across the run’s stored layers at one turn token. The x axis lists those layers at even spacing. The fan widens with depth in every family.

A.1 The ordinal gradient at the turn tokens

We fit the principal component separately at each layer and each turn token, then rank-correlate it with the stated horizon. Table A.2 gives the strongest correlation over layers for each token. The component is fit independently per cell, so its sign carries no meaning and the table reports magnitude.

Model	Domain	Turn token	$ \rho $
gemma-2-9b-it	climate	<end_of_turn>	0.965
		\n	0.963
		model	0.927
Qwen3-4B-Instruct-2507	startup	< im_end >	0.924
		< im_start >	0.918
		\n	0.855
Llama-3.1-8B-Instruct	health	\n\n	0.919
		< end_header_id >	0.845
		< start_header_id >	0.843
Mistral-7B-Instruct-v0.3	education	[/INST]	0.901
Qwen3-4B-Instruct-2507	investment	< im_start >	0.892
		< im_end >	0.842
		assistant	0.792

Table A.2: Rank correlation between the first principal component of `resid_post` and the stated time horizon, at the turn-transition tokens. Each row gives the strongest layer for that token. Horizons span seconds to centuries.

A reward control. These prompts sweep reward on a log grid alongside the horizon, so the same activations carry a second graded quantity. We correlate the same component with reward and compare.

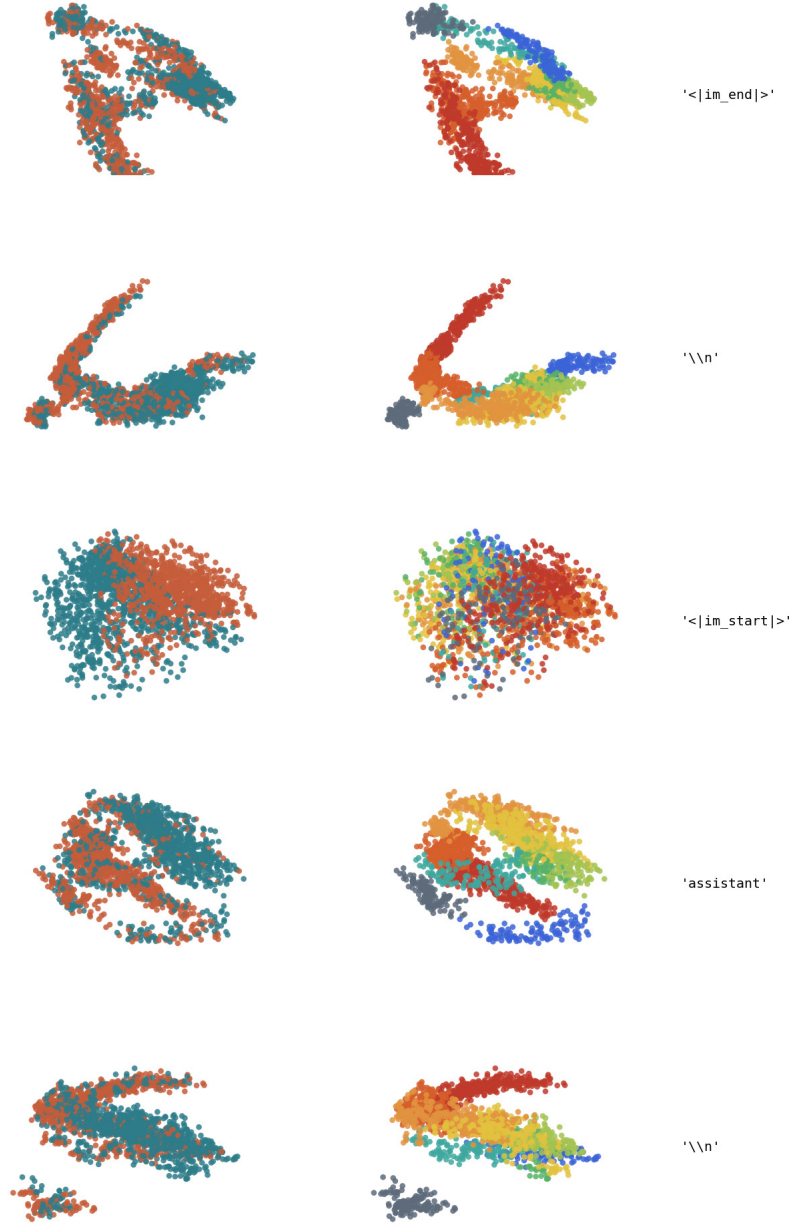


Figure A.1: Three components of `resid_post` at the turn tokens of the investment run, at L21. The left column colors by preference and the right by horizon. The choices stay mixed while the horizon stays ordered.

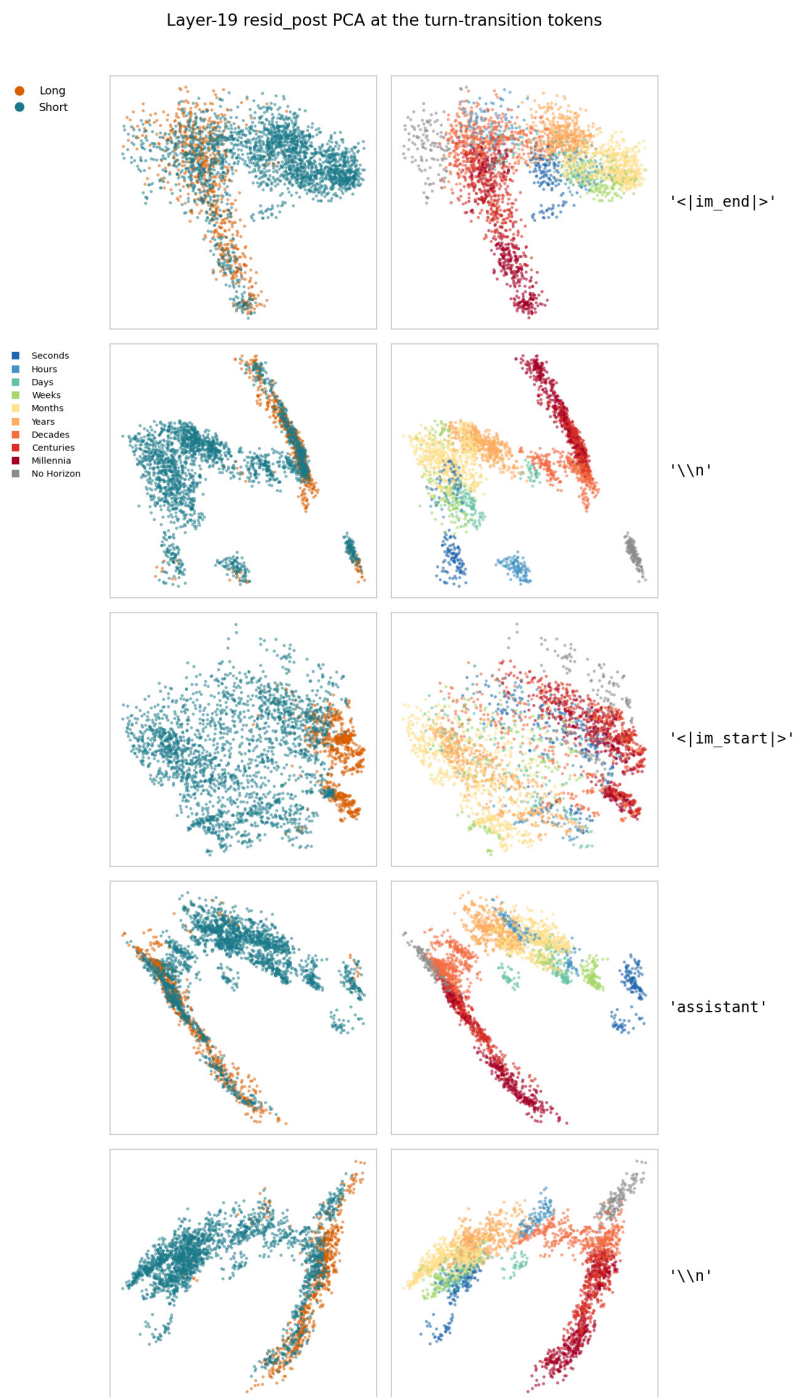


Figure A.2: Qwen3-4B-Instruct-2507 on the startup domain, resid_post at L19 of 36. The target model on a second domain, with the same ordering.

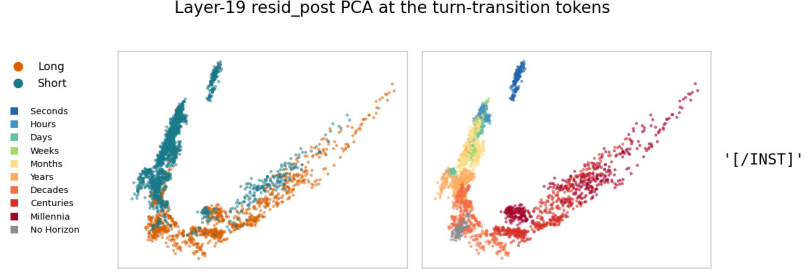


Figure A.3: Mistral-7B-Instruct-v0.3 on the education domain, `resid_post` at L19 of 32. Mistral’s template carries a single turn token, `[/INST]`, so its turn window is one position wide.

Model	Domain	Horizon $ \rho $	Reward $ \rho $
gemma-2-9b-it	climate	0.965	0.792
Qwen3-4B-Instruct-2507	startup	0.924	0.719
Llama-3.1-8B-Instruct	health	0.919	0.223
Mistral-7B-Instruct-v0.3	education	0.901	0.627
Qwen3-4B-Instruct-2507	investment	0.892	0.495

Table A.3: The strongest ordering of each quantity, over layers and turn tokens, in the same activations. Horizon orders the component more strongly than reward in every family.

We therefore posit that the ordering belongs to the horizon rather than to graded quantities in general.

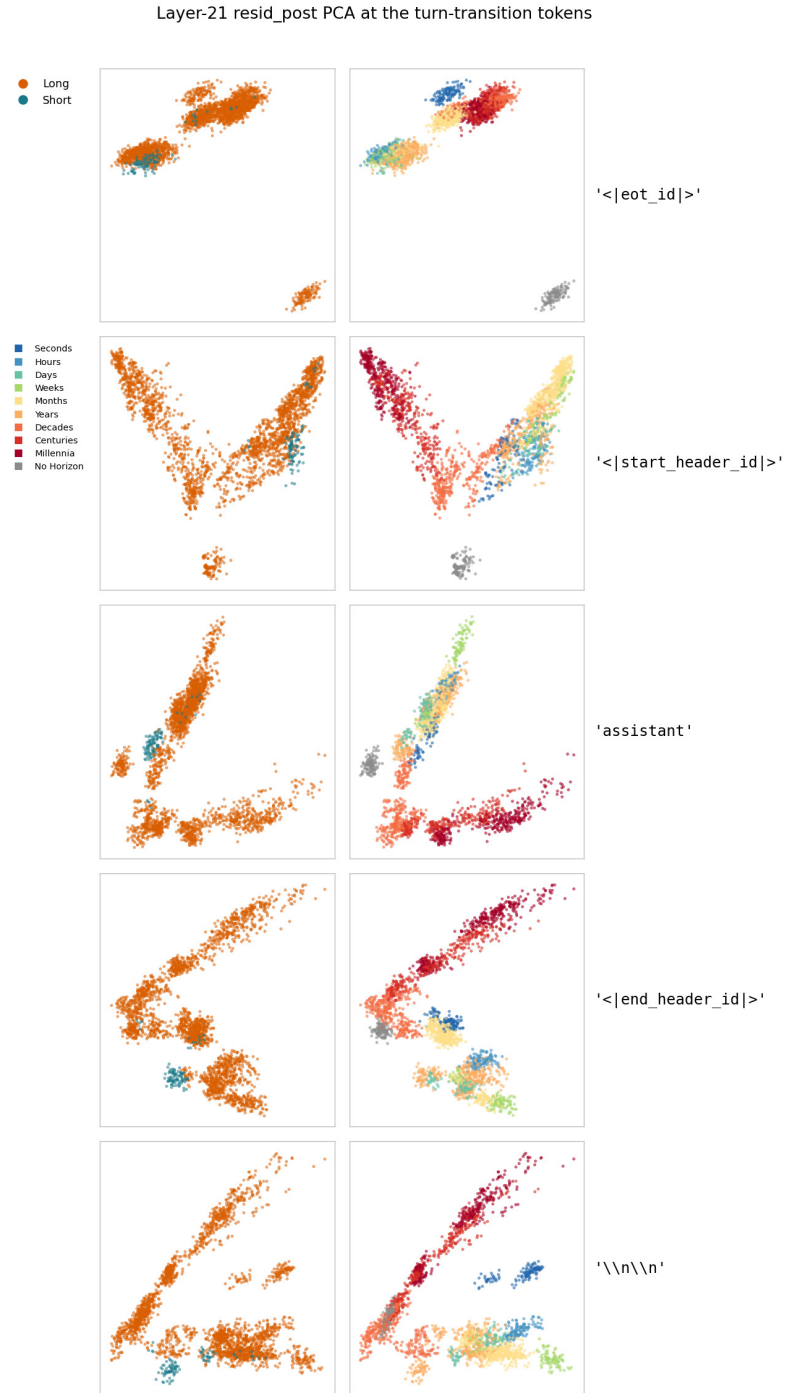


Figure A.4: Llama-3.1-8B-Instruct on the health domain, `resid_post` at L21 of 32. Rows are the five turn-transition tokens, ending at the double newline that opens the response. The left column is colored by preference and the right by horizon.

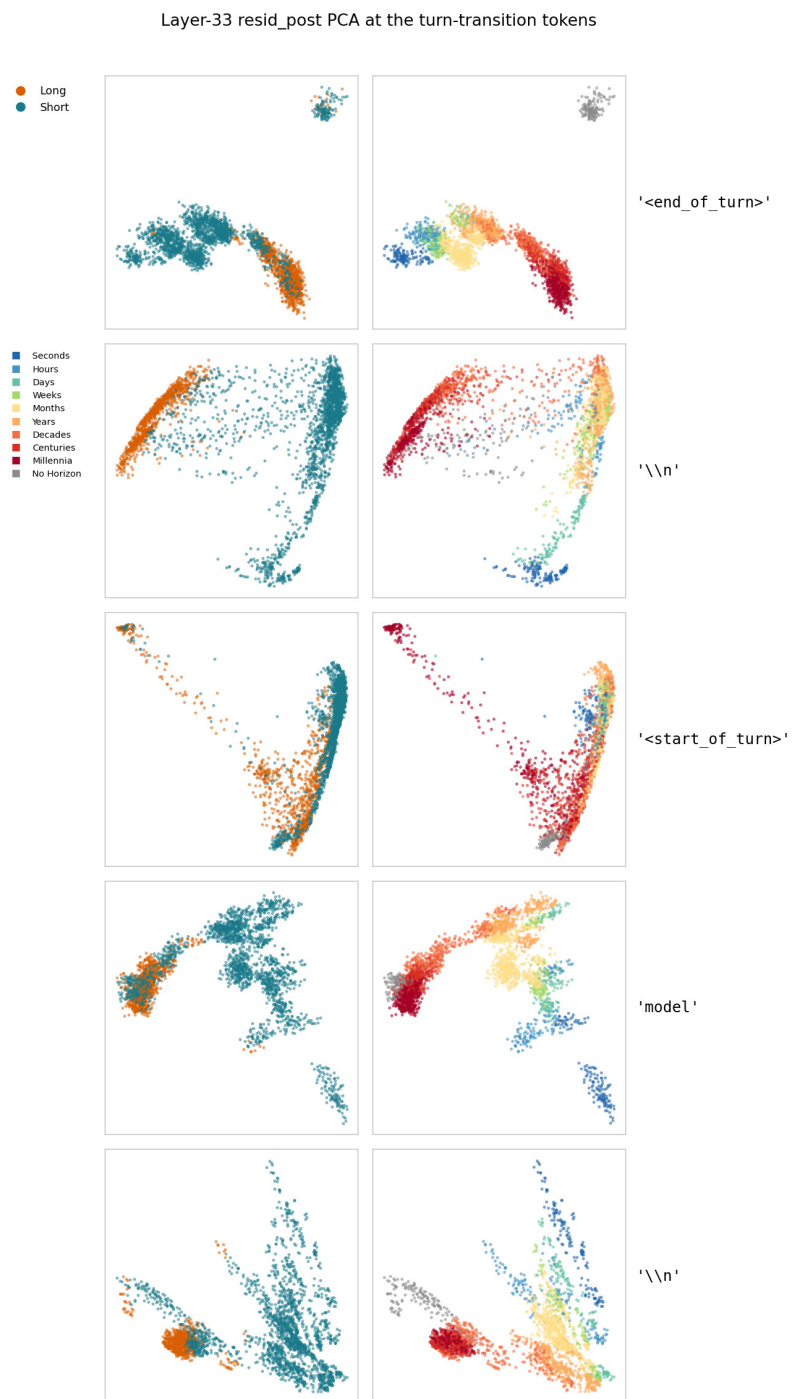


Figure A.5: `gemma-2-9b-it` on the climate domain, `resid_post` at L33 of 42. Rows are Gemma's own turn tokens, ending at the newline that opens the response.

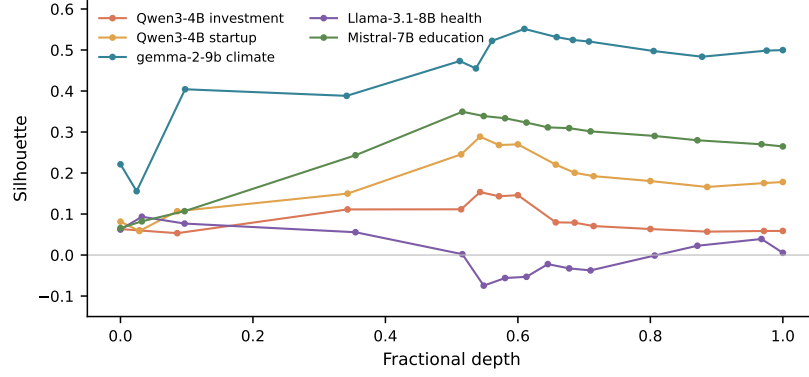


Figure A.6: Silhouette of the two preference classes at the final turn token, over each run’s stored layers at fractional depth. Separation is partial at best and varies by family. The shared structure at these tokens is the horizon gradient.

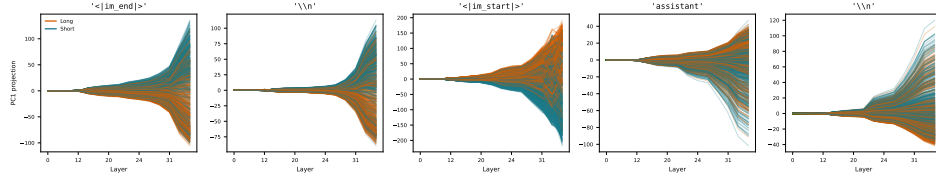


Figure A.7: PC1 fans for Qwen3-4B-Instruct-2507 on the investment domain.

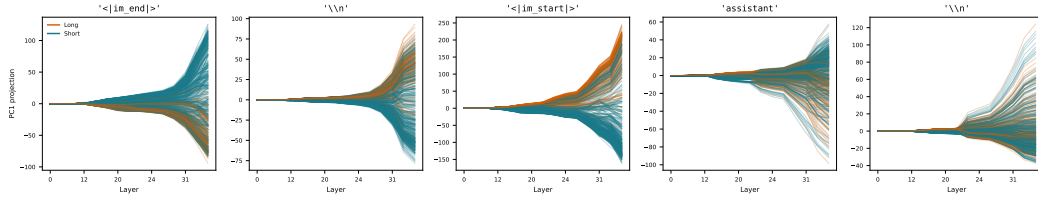


Figure A.8: PC1 fans for Qwen3-4B-Instruct-2507 on the startup domain.

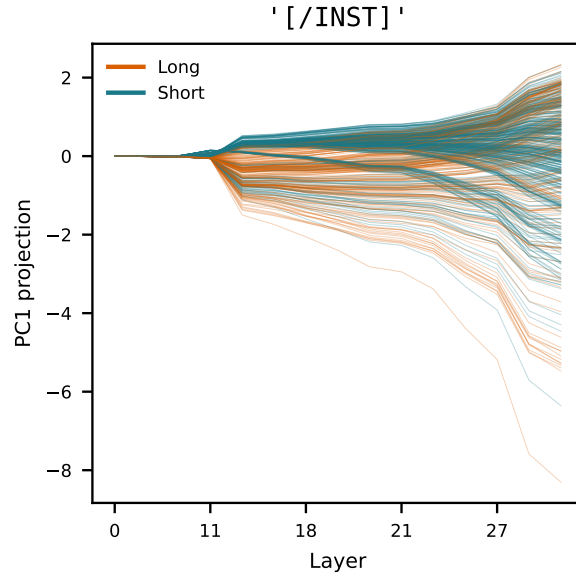


Figure A.9: PC1 fans for Mistral-7B-Instruct-v0.3 on the education domain, at its single turn token.

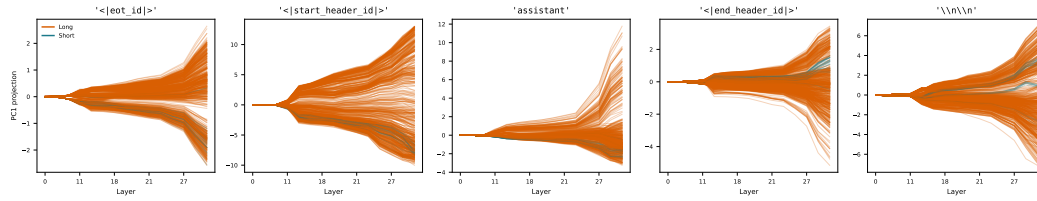


Figure A.10: PC1 fans for Llama-3.1-8B-Instruct on the health domain, one panel per turn token.

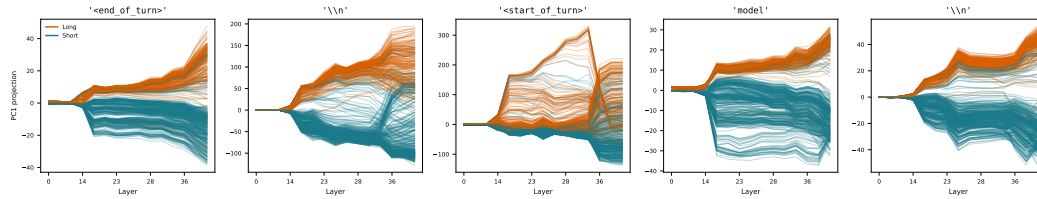


Figure A.11: PC1 fans for gemma-2-9b-it on the climate domain.

Appendix B Contrastive steering results

Localization and geometry describe where temporal preference sits and what shape it has. Neither shows that changing it changes the model’s choice. We build a **contrastive activation vector**, the difference between the mean activation on long-term prompts and the mean activation on immediate prompts. We add it at a candidate layer during inference, scaled by a coefficient, and measure what the model then chooses.

Controls. We score every configuration against a random direction of matched norm at the same layer. We counterbalance the order of the two options in the forced-choice prompt. We index layers by fractional depth, so the same sweep means the same thing in models of different depth.

Vector recipe. A variant derives the vector as the mean of teacher-forced label activations at the best probe layer, rather than as a unit-normalized difference of means. This variant also steers `gemma-2-9b-it` at larger magnitude ($\alpha = 150$ at L25), so the effect is robust to the vector recipe.

B.1 Sweeps per model

The sweep covers fractional depths 0.50 to 0.65 and four coefficients. The best cell for each model appears in the main text.

α	0.50	0.55	0.58	0.61	0.65
10	3.9	3.1	2.8	2.1	0.9
20	5.9	3.7	3.9	2.9	0.9
35	4.7	-0.2	3.3	3.8	0.8
50	4.6	1.1	1.4	3.6	0.6

Table B.1: Forced-choice score for `Qwen3-4B-Instruct-2507` over fractional depth and coefficient. An underlined cell is one where the vector does not beat the matched-norm random direction.

α	0.50	0.55	0.58	0.61	0.65
10	<u>1.4</u>	2.4	2.8	2.5	2.2
20	12.1	4.8	5.8	3.3	2.9
35	13.2	17.4	14.1	7.7	<u>5.1</u>
50	11.2	11.5	8.7	7.4	5.8

Table B.2: Forced-choice score for `Llama-3.1-8B-Instruct` over fractional depth and coefficient. An underlined cell is one where the vector does not beat the matched-norm random direction.

α	0.50	0.55	0.58	0.61	0.65
10	2.2	1.7	1.6	1.4	1.4
20	2.7	2.0	1.9	1.5	1.5
35	3.4	2.4	2.2	1.6	1.6
50	3.9	2.7	2.5	1.7	1.6

Table B.3: Forced-choice score for `gemma-2-9b-it` over fractional depth and coefficient. An underlined cell is one where the vector does not beat the matched-norm random direction.

α	0.50	0.55	0.58	0.61	0.65
10	5.4	-0.7	-1.2	-1.3	-1.5
20	10.7	<u>6.4</u>	12.1	5.4	2.6
35	10.2	8.4	8.0	6.3	7.3
50	10.3	9.0	8.3	6.9	7.3

Table B.4: Forced-choice score for **Mistral-7B-Instruct-v0.3** over fractional depth and coefficient. An underlined cell is one where the vector does not beat the matched-norm random direction.

What the steered text looks like. Under a negative coefficient the model advises immediate communication and asks what the deadline is. Under a positive coefficient the same prompt yields resilience, shared purpose, and adaptive capacity. The answer stays on topic.

Where the vector stops working. Above the swept range the output loses coherence. The vector is a straight line and the representation is curved, so a large enough step leaves the region where the line approximates the manifold. We therefore conjecture that steering along the curvature would buy the same shift without the loss of quality.

B.2 Side effects on instruction following

A steering vector is only usable if it leaves the rest of the model’s competence intact. We steer each model in the long-term direction and score it on IFEval [58], before and after the intervention (Table B.5). Moderate-strength steering costs three to five points of instruction-following accuracy, and the cost is similar across the two families we test.

Model		Strict		Loose	
		Prompt	Instruction	Prompt	Instruction
Qwen3-4B-Instruct-2507 ($\alpha = 50$, $N = 464$)	baseline	0.705	0.786	0.722	0.799
	steered	0.659	0.752	0.681	0.766
gemma-2-9b-it ($\alpha = 150$, $N = 510$)	baseline	0.712	0.790	0.730	0.804
	steered	0.676	0.757	0.692	0.772

Table B.5: IFEval accuracy before and after steering in the long-term direction with the contrastive vector. N is the number of scored prompts after skipping malformed responses (77 for **Qwen3-4B-Instruct-2507**, 31 for **gemma-2-9b-it**).